YISAK MESIFIN

Cloud Engineer

📧 yisakmesifin@gmail.com · 🔗 linkedin.com/in/yisak-mesifin-815946354 (https://linkedin.com/in/yisak-mesifin-815946354) · 💻 github.com/yisakm9 https://github.com/yisakm9 · 🌐 yisakmesifin.org · 📍 Ethiopia

SUMMARY

Cloud Engineer with hands-on experience designing, deploying, and securing production-grade infrastructure on AWS and Google Cloud Platform. Proficient in Infrastructure as Code (Terraform), CI/CD automation (GitHub Actions), containerization (Docker/Cloud Run), and DevSecOps practices. Built and deployed 4+ end-to-end cloud projects featuring serverless architectures, event-driven pipelines, AI/ML integrations, real-time monitoring, and security-hardened infrastructure — each with modular Terraform, automated CI/CD, and comprehensive documentation.

TECHNICAL SKILLS

Domain | Technologies

Cloud Platforms AWS (Lambda, S3, DynamoDB, API Gateway, CloudFront, SQS, SNS, SES, EventBridge, CloudWatch, KMS, IAM, Cost Explorer, Transcribe, Bedrock, Rekognition) · GCP (Cloud Run, Cloud SQL, VPC, Load Balancer, Cloud NAT, Cloud Monitoring, Secret Manager, Artifact Registry, IAM)

Infrastructure as Code Terraform (modular architecture, remote state, multi-environment, lifecycle management)

CI/CD & DevSecOps GitHub Actions (multi-stage pipelines, OIDC/WIF keyless auth) · TFLint · Checkov · Trivy

Containers & Compute Docker (multi-stage builds) · Cloud Run · Lambda · Serverless architecture |

Networking VPC design · Private subnets · Cloud NAT · VPC peering · Serverless VPC Connectors · Firewall rules · Global HTTP(S) Load Balancers · CloudFront CDN

Databases Cloud SQL PostgreSQL · DynamoDB · SQLAlchemy ORM

Monitoring & Observability Cloud Monitoring (custom metrics, alert policies, uptime checks) · CloudWatch (alarms, logs, events) · SNS alerting

Security Workload Identity Federation (OIDC) · KMS encryption · Secret Manager · Least-privilege IAM · VPC Flow Logs · Non-root containers

Languages & Frameworks Python (FastAPI, Boto3, SQLAlchemy, pytest)

PROJECTS

☁️ CloudPulse — Cloud-Native Uptime Monitoring Platform

Google Cloud Platform · [GitHub](https://github.com/yisakm9/CloudPulse-Uptime-Monitor)

- A production-grade, serverless uptime monitoring service that continuously monitors website and API health, tracks response times, calculates uptime percentages, and sends automated alerts when services go down.

- Architected a multi-service serverless platform using Cloud Run (web dashboard + background worker), Cloud SQL PostgreSQL (private IP), and a Global HTTP(S) Load Balancer with a static IP entry point

- Designed a 59-resource infrastructure across 12 modular Terraform modules (VPC, NAT, Firewall, Cloud SQL, Cloud Run, Load Balancer, IAM, Monitoring), enabling full destroy-and-rebuild in under 15 minutes

- Implemented a 4-stage CI/CD pipeline (validate → infrastructure → build → deploy) using GitHub Actions with Workload Identity Federation** (keyless OIDC authentication — zero secrets stored)

- Built a real-time alerting system using Cloud Monitoring custom metrics : worker detects endpoint DOWN → writes custom metric → alert policy fires → email notification within 60 seconds

- Enforced defense-in-depth security: private database (no public IP), Secret Manager for credentials, firewall deny-all default , least-privilege IAM (3 dedicated service accounts), VPC Flow Logs, and non-root Docker containers

- Created 18 automated unit tests (pytest) covering API health checks, CRUD operations, and uptime calculation logic

Tech: Cloud Run · Cloud SQL PostgreSQL · VPC · Cloud NAT · Global LB · Cloud Monitoring · Secret Manager · Artifact Registry · Terraform · GitHub Actions · WIF · Docker · FastAPI · SQLAlchemy · pytest

💰 Automated AWS Cost Calculator & Alerting System

Amazon Web Services · [GitHub](https://github.com/yisakm9/Project-12-Cloud-Cost-Calculator)

> An end-to-end serverless cost monitoring platform that provides real-time AWS cost visibility through a dynamic dashboard and proactive budget alerting — solving the "surprise cloud bill" problem.

- Architected a serverless cost monitoring system using Lambda, API Gateway, CloudFront, and S3 , providing at-a-glance weekly cost breakdowns by AWS service through a dynamic web dashboard

- Implemented automated budget alerting via CloudWatch Billing Alarms → SNS, enabling proactive cost intervention before threshold breaches — plus automated weekly HTML cost reports via SES

- Hardened infrastructure security using Checkov static analysis , KMS encryption (at-rest and in-transit), CloudFront OAI (no direct S3 access), and least-privilege IAM roles for every Lambda function

- Automated end-to-end deployment with a GitHub Actions CI/CD pipeline integrating TFLint linting, Checkov security scanning, Terraform plan/apply, and automatic frontend deployment to S3

- Designed the infrastructure as 9 reusable Terraform modules (Lambda, API Gateway, CloudFront, S3, CloudWatch, SNS, SES, IAM, KMS) for reproducible, version-controlled deployments

Tech: Lambda · API Gateway · CloudFront · S3 · CloudWatch · SNS · SES · KMS · IAM · Cost Explorer · Terraform · GitHub Actions · OIDC · Checkov · TFLint

🔄 Advanced Event-Driven S3 Backup System

Amazon Web Services · [GitHub](https://github.com/yisakm9/Project-13-Automatic-Backup-System)

> A professional-grade, automated disaster recovery system that replicates S3 objects across AWS regions, validates backup integrity via checksum comparison, and alerts operators on failures — all without human intervention.

- Designed a fully event-driven backup pipeline using S3 Events → EventBridge → SQS → Lambda, eliminating cron jobs and delivering zero-downtime, hands-free cross-region replication for 3 data categories (documents, media, database)

- Built a data integrity validation system : a Lambda function automatically compares checksums (ETags) between primary and replica objects, with automatic retry logic to handle S3 replication lag gracefully

- Implemented resilience engineering patterns: SQS buffering for backpressure management, Dead-Letter Queues (DLQs) for poison-pill message isolation, and a failure-notifier Lambda → SNS alerting chain for operator visibility

- Enforced end-to-end encryption across all resources using Customer-Managed KMS Keys (S3 buckets, SQS queues, SNS topics, Lambda environment variables) with zero public access on all storage

- Integrated DevSecOps into the CI/CD pipeline with TFLint and Checkov static analysis, preventing security misconfigurations from reaching production

Tech: S3 (CRR) · EventBridge · SQS · Lambda · SNS · KMS · IAM · DLQ · Terraform · GitHub Actions · OIDC · Checkov · TFLint

🤖 The Meeting Analyst — AI-Powered Transcription & Summarization Pipeline

Amazon Web Services· [GitHub](https://github.com/yisakm9/The-Meeting-Analyst)

> An end-to-end AI pipeline that automatically transcribes meeting recordings using Amazon Transcribe and generates intelligent summaries with key decisions and action items using Generative AI (Amazon Bedrock / Meta Llama 3).

- Architected an asynchronous, event-driven AI pipeline : audio upload to S3 triggers SQS → Lambda → Transcribe → EventBridge → Lambda → Bedrock (Llama 3) → DynamoDB, processing meetings from raw audio to structured summaries without human intervention

- Integrated Amazon Bedrock (Meta Llama 3 foundation model) to extract concise summaries, key decisions, and action items from meeting transcripts — demonstrating practical Generative AI integration in cloud workflows

- Built a serverless REST API using API Gateway → Lambda → DynamoDB for on-demand retrieval of meeting summaries by ID, with JSON response formatting

- Designed the system for fault tolerance: SQS decouples ingestion from processing, EventBridge handles async job completion events, and SNS delivers completion notifications to stakeholders

- Managed the entire infrastructure through modular Terraform (S3, IAM, Lambda, SQS, EventBridge, DynamoDB, API Gateway, SNS) with GitHub Actions CI/CD deployment

Tech: Transcribe · Bedrock (Llama 3) · Lambda · S3 · SQS · EventBridge · DynamoDB · API Gateway · SNS · IAM · Terraform · GitHub Actions

CERTIFICATIONS

Certification Status

AWS Solutions Architect – Associate In Progress

HashiCorp Terraform Associate Planned

Google Cloud Associate Cloud Engineer Planned

> Update the Status column as you earn these certifications

EDUCATION

Bachelor of Science in Information Technology (3rd Year Student)

Mekelle University · Expected Graduation: 2027

ADDITIONAL

- Multi-Cloud: Demonstrated proficiency across both AWS (4 projects) and GCP (1 project) with Terraform as a unified IaC layer

- Documentation: Every project includes professional Architecture Decision Records (ADRs), operations runbooks, and Mermaid architecture diagrams

- Open Source: All projects publicly available with comprehensive README documentation on GitHub